Options IV: Binomial Trees

BUSS386. Futures and Options

Prof. Ji-Woong Chung

Lecture Outline

- A real option quote, and the question of who decided the price — today's lecture is the answer
- One step: build the option out of stock and bond, and the price falls out — no probabilities needed
- Risk-neutral valuation: the same price by a shortcut, and why DCF gives the same answer the hard way
- Two steps, then N: backward induction, and a hedge that has to be rebalanced
- American options: at every node, exercise or wait
- Choosing u and d from volatility, other underlyings, and a detour into corporate finance

Who decided this price?

- Lecture 7 gave us **bounds**: at least $17.05 - 17e^{-rT} = \$0.19$, at most \$17.05. The market says \$1.26 to \$1.45. Something sharper is at work.
- The market maker is not guessing. She has a **recipe**.
- Today: the recipe on the crudest model there is. Each step, the stock goes up or down, nothing else.
- Next week the steps shrink to nothing and we price this very call.

Coupang call, close of 4 Nov 2026	
Stock price S_0	\$17.05
Strike K	\$17
Expiry	15 Jan 2027
Days to expiry	72
Call bid	\$1.26
Call ask	\$1.45
3-month T-bill	4.2%

[Yahoo Finance] [Cboe quotes] [US Treasury]

Today: **build an option out of stock and bond**. Its price is what the ingredients cost — and the recipe tells the hedger what to hold.

Why we need a model at all

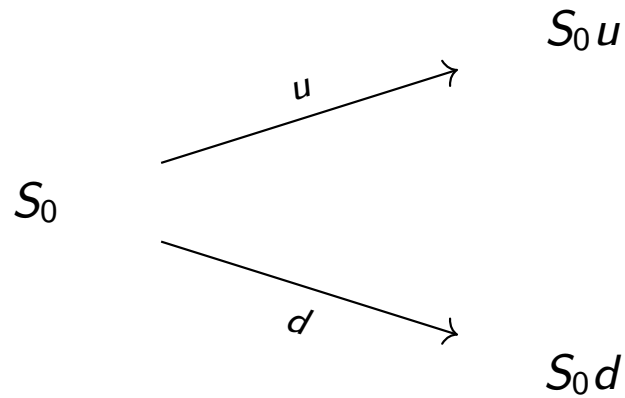
- Lecture 7 gave us **bounds** and **parity**. Useful, but not a price.
- The textbook DCF route, $e^{-r_{\text{call}}T} \times E[\max(S_T - K, 0)]$, needs two things we do not have: the distribution of S_T and the discount rate r_{call}.
- The discount rate is the real problem. A call is a **levered** position in the stock, and the leverage changes every time the stock moves — so r_{call} changes too.
- The binomial tree sidesteps both: a model of S_T simple enough to draw, and a **no-arbitrage** argument that never asks for r_{call}.

Inputs the tree will use: S_0, K, T, volatility, r, and dividends — the same six that drive every option price.

One-Step Binomial Tree

The setting

- Over one time step the stock price can do **only two things**: go up by a factor $u > 1$ or down by a factor $d < 1$.
- There is no arbitrage.
- That is all. We do not say how *likely* up is.

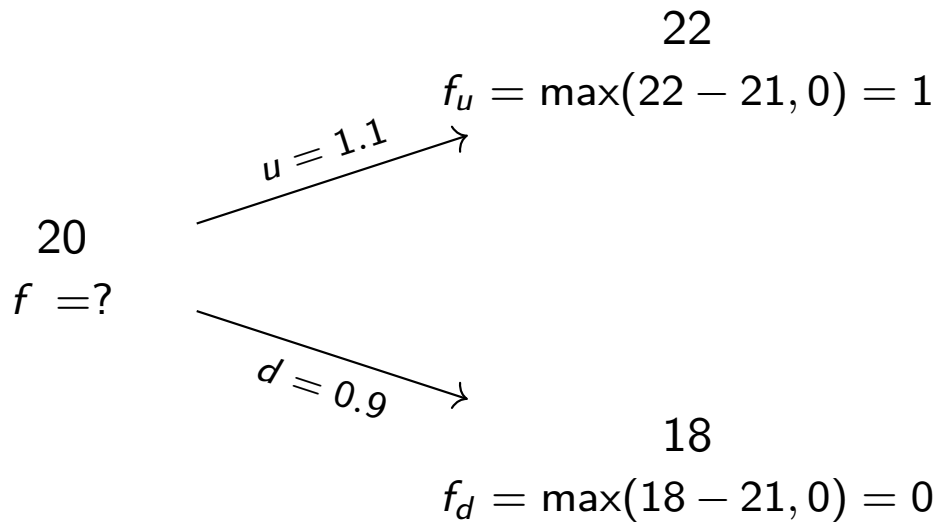


- Two branches look absurdly crude. With many short steps the tree becomes as fine as you like.

Example — a call on a two-branch stock

e.g. Stock at \$20. In three months it will be either \$22 or \$18. A 3-month European call has $K = 21$; $r = 12\%$ (c.c.).

- **Step 1.** Write the option's payoff at each end node.



- What is the call worth today?

Step 2 — replicate the payoff with stock and bond

- Buy x shares and hold a bond paying y dollars at T ($y < 0$ means borrowing). Choose x and y so the portfolio pays *exactly* what the call pays in *both* states:

$$\begin{cases} 22x + y = 1 & \text{(up)} \\ 18x + y = 0 & \text{(down)} \end{cases} \qquad \Longrightarrow \qquad x = 0.25, \quad y = -4.5.$$

- Recipe: **buy 0.25 shares** and **borrow** $4.5e^{-0.12\times 0.25} = \$4.37$ today, repaying \$4.50 in three months.

At T	$S_T = 22$	$S_T = 18$
0.25 shares	5.50	4.50
Repay loan	-4.50	-4.50
Portfolio	1.00	0.00
Call	1.00	0.00

Step 3 — the price is what the recipe costs

- Two portfolios with identical payoffs in every state must have the same price today, or there is an arbitrage. So

$$f \;=\; S_0 x + y e^{-rT} \;=\; 20 \times 0.25 - 4.5\, e^{-0.12\times 0.25} \;=\; \boxed{\$0.633}.$$

- We never used the **probability** of an up move. It is not needed — and not because it does not matter: it is already baked into $S_0 = 20$.

- Nor did we use anyone's **risk preference**. Two investors who disagree about the stock's prospects must agree on this call price, given S_0.

What if the call trades at \$0.70?

- Then the call is **rich** relative to its recipe. Sell the expensive thing, buy the cheap copy:

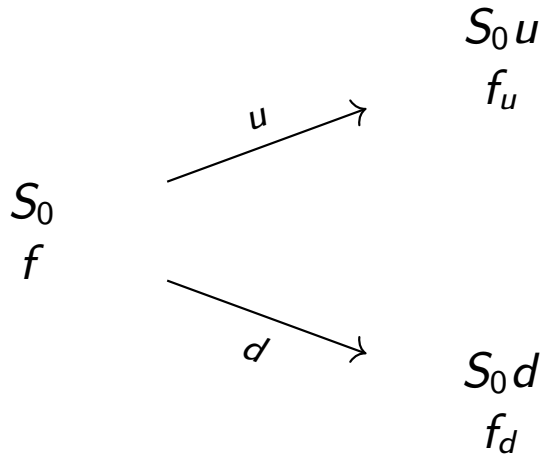
	Today	$S_T = 22$	$S_T = 18$
Sell one call	$+0.700$	-1.00	0
Buy 0.25 shares	-5.000	$+5.50$	$+4.50$
Borrow $4.5e^{-rT}$	$+4.367$	-4.50	-4.50
Net	$\boxed{+0.067}$	0	0

- Pocket 6.7 cents per call today, owe nothing in either state. At \$0.55 run it backwards: buy the call, short 0.25 shares, lend.

The general one-step formula — replication

- Payoffs: $f_u = \max(S_0 u - K, 0)$, $f_d = \max(S_0 d - K, 0)$.
- Replicate: $S_0 u\, x + y = f_u$ and $S_0 d\, x + y = f_d$, so

$$x = \frac{f_u - f_d}{S_0 u - S_0 d}, \qquad y = \frac{u f_d - d f_u}{u - d}.$$



- Price: $f = S_0 x + y e^{-rT}$.
  - x is the **hedge ratio**: the change in option value per \$1 change in the stock. Lecture 11 will call it **delta** and build a whole chapter on it.

The general one-step formula — rearranged

- Substitute x and y into $S_0 x + y e^{-rT}$ and collect the f_u and f_d terms:

$$f = \frac{f_u - f_d}{u - d} + e^{-rT}\,\frac{u f_d - d f_u}{u - d} = e^{-rT}\left[\frac{e^{rT} - d}{u - d}\,f_u + \frac{u - e^{rT}}{u - d}\,f_d\right]$$

- Define $p = \dfrac{e^{rT} - d}{u - d}$. Then

$$\boxed{f = e^{-rT}\left[\,p\,f_u + (1 - p)\,f_d\,\right]}$$

- p lies between 0 and 1 exactly when $d < e^{rT} < u$ — which no-arbitrage requires anyway (otherwise the stock would dominate the bond, or vice versa).

Example — the same call, by formula

- $u = 22/20 = 1.1$, $d = 18/20 = 0.9$, $r = 12\%$, $T = 0.25$:

$$p = \frac{e^{0.12 \times 0.25} - 0.9}{1.1 - 0.9} = 0.652.$$

- Price:

$$f = e^{-0.12 \times 0.25}\big[\,0.652 \times 1 + 0.348 \times 0\,\big] = \boxed{\$0.633}.$$

- Same number as the recipe — as it must be: the formula *is* the recipe, rearranged.

Risk-Neutral Valuation

Reading the formula as a probability

- $e^{-rT}[\,pf_u + (1-p)f_d\,]$ looks like "expected payoff, discounted" — a DCF — with p playing the role of a probability.
- Two things are odd. The discount rate is the **risk-free** rate, although the payoff is risky. And p was never estimated from data; it came from u, d and r.
- Check what p does to the stock: $pS_0u + (1-p)S_0d = S_0e^{rT}$. Under p, the stock is expected to earn the **risk-free rate**.
- So p is the probability that would be right *if* investors were risk-neutral. We call it the **risk-neutral probability**, and the method **risk-neutral valuation**.

Risk-neutral valuation vs. DCF

- Let p^* be the *real* probability of an up move, the one you would estimate from history.

	Risk-neutral valuation	DCF
Probability of up	$p = \dfrac{e^{rT} - d}{u - d}$	p^* (from data)
Discount rate	r	r_{call} (from a model)
Price	$e^{-rT}[\, pf_u + (1-p)f_d \,]$	$e^{-r_{\text{call}}T}[\, p^* f_u + (1-p^*)f_d \,]$

- Both columns must give the **same price** — there is only one no-arbitrage price. Let us check by doing the DCF the hard way.

DCF, step 1 — the real probability

- Suppose investors require the stock to earn α per annum in the real world. The real probability must deliver that expected return:

$$p^* S_0 u + (1 - p^*) S_0 d = S_0 e^{\alpha T} \qquad \Longrightarrow \qquad p^* = \frac{e^{\alpha T} - d}{u - d}.$$

- Same formula as p, with α in place of r. If $\alpha > r$ then $p^* > p$.

DCF, step 1 — the real probability

Q1. In the example ($20 \to 22$ or 18 in three months), investors require $\alpha = 16\%$ per annum on the stock. What is the real probability p^* of an up move?

DCF, step 1 — the real probability

Q1. $\alpha = 16\%$, $u = 1.1$, $d = 0.9$, $T = 0.25$. Real probability?

A.

$$p^* = \frac{e^{0.16 \times 0.25} - 0.9}{1.1 - 0.9} = \boxed{0.704} \quad \text{versus} \quad p = 0.652.$$

- The real world is more optimistic about the stock than the risk-neutral world. We will see why that is exactly right.

DCF, step 2 — the discount rate for the call

- The call *is* a portfolio of x shares and a bond. So its required return is the **value-weighted average** of the stock's return α and the bond's return r:

$$e^{r_{\text{call}}T} = w_S\, e^{\alpha T} + w_B\, e^{rT}, \qquad w_S = \frac{S_0 x}{S_0 x + y e^{-rT}}, \quad w_B = \frac{y e^{-rT}}{S_0 x + y e^{-rT}}.$$

- With $y < 0$ the bond weight is negative: the call is stock bought **on margin**, and a lot of it.

DCF, step 2 — the discount rate for the call

Q2. Same call ($K = 21$, 3 months, $r = 12\%$, $\alpha = 16\%$). What discount rate r_{call} does DCF require?

DCF, step 2 — the discount rate for the call

Q2. Discount rate for the call?

A. The recipe was $x = 0.25$ shares ($5.00) and a loan of $4.37, net $0.633:

$$e^{r_{\text{call}} \times 0.25} = \frac{5.00}{0.633}\, e^{0.16 \times 0.25} + \frac{-4.37}{0.633}\, e^{0.12 \times 0.25}$$
$$= 7.90 \times 1.0408 - 6.90 \times 1.0305 = 1.1123$$

so $r_{\text{call}} = 4 \ln(1.1123) = \boxed{42.6\%}$ per annum.

- Weights of $+790\%$ and -690%: this call is a stock position levered almost eight times. That is why its required return is so far above the stock's 16%.
- And the leverage changes with S_0. Price the same call with the stock at $21 and r_{call} changes. DCF is exact here only because the tree told us x and y.

DCF, step 3 — the price

Q3. Using p^* and r_{call}, price the call by DCF. Does it match the risk-neutral price of \$0.633?

DCF, step 3 — the price

Q3. DCF price?

A.

$$e^{-0.426\times 0.25}\big[\,0.704\times 1+0.296\times 0\,\big]=0.899\times 0.704=\boxed{\$0.633}.$$

- Identical. The higher probability of the good state (0.704 vs 0.652) is exactly undone by the higher discount rate (42.6% vs 12%).

Two routes, one price

- Risk aversion can be charged in **two places**: in the discount rate (DCF: $r_{\text{call}} > r$) or in the probabilities (risk-neutral: $p < p^*$, more weight on the bad state).
- Risk-neutral valuation is an **interpretation** of the replication argument, not a claim that anyone is risk-neutral.
- It wins on convenience: r is observable; α and r_{call} are not, and r_{call} changes at every node.
- This is why every pricing model from here on — trees, Black–Scholes, Monte Carlo — is written in the risk-neutral world.

Two Steps, then N

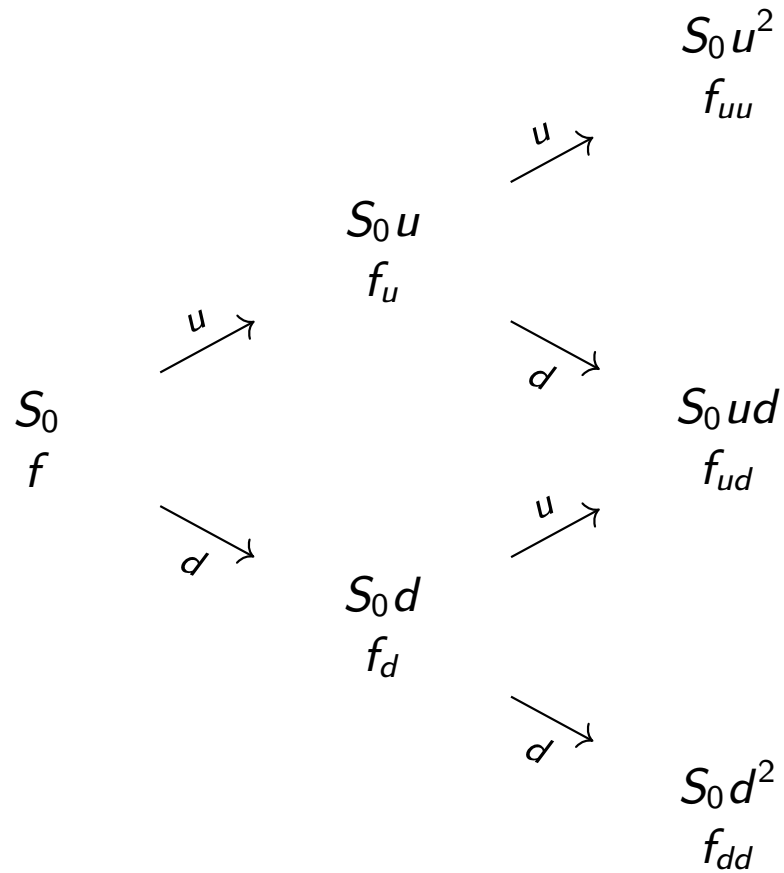
Two-step tree — work backwards

- Each step has length Δt; u, d, r as before; the option expires after two steps.
- Each node at Δt sits on top of a **one-step tree**, so the one-step formula applies there:

$$f_u = e^{-r\Delta t}[\, pf_{uu} + (1-p)f_{ud}\,]$$
$$f_d = e^{-r\Delta t}[\, pf_{ud} + (1-p)f_{dd}\,]$$

with $p = \dfrac{e^{r\Delta t} - d}{u - d}$.

- Then once more at time 0:
  $f = e^{-r\Delta t}[\, pf_u + (1-p)f_d\,]$.



The tree **recombines**: up-then-down and down-then-up meet at S_0ud. Three end nodes, not four.

Two-step tree — collapsed to one line

- Substitute f_u and f_d into the time-0 formula:

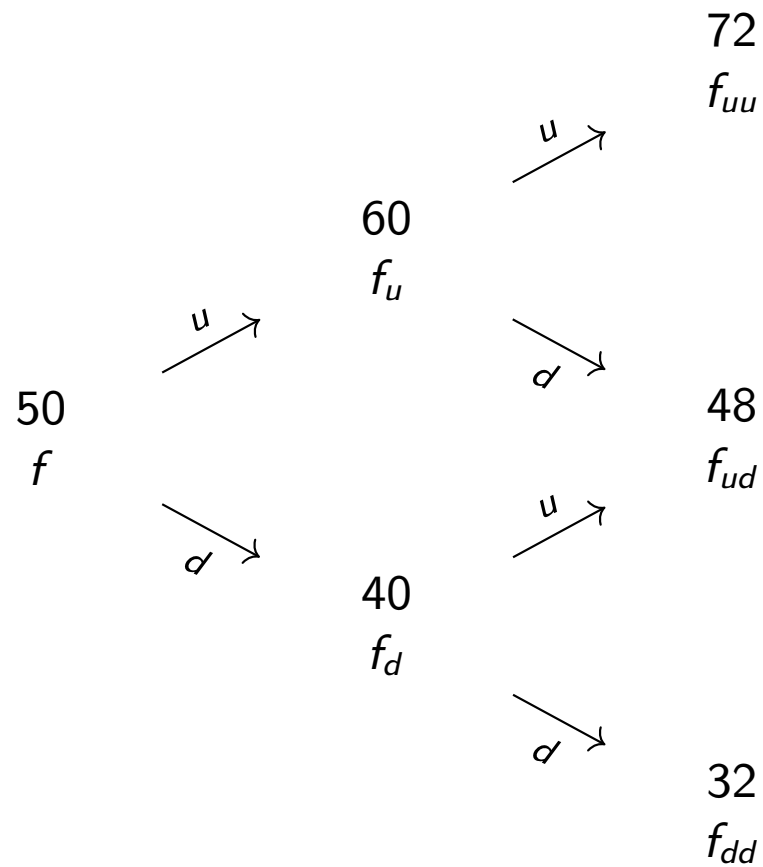
$$f = e^{-2r\Delta t}\Big[\, p^2 f_{uu} + 2p(1-p)\, f_{ud} + (1-p)^2 f_{dd} \,\Big].$$

- Read it as a risk-neutral expectation: p^2, $2p(1-p)$ and $(1-p)^2$ are the risk-neutral probabilities of the top, middle and bottom end nodes.

- The 2 counts **paths**: there are two ways to reach the middle node.

Example — a two-step European put

Q. A 2-year European put has $K = 52$ on a stock at $S_0 = 50$. Each year the stock moves up or down by 20%; $r = 5\%$ (c.c.). What is the put worth?



Example — a two-step European put

A. $p = \dfrac{e^{0.05} - 0.8}{1.2 - 0.8} = 0.6282.$

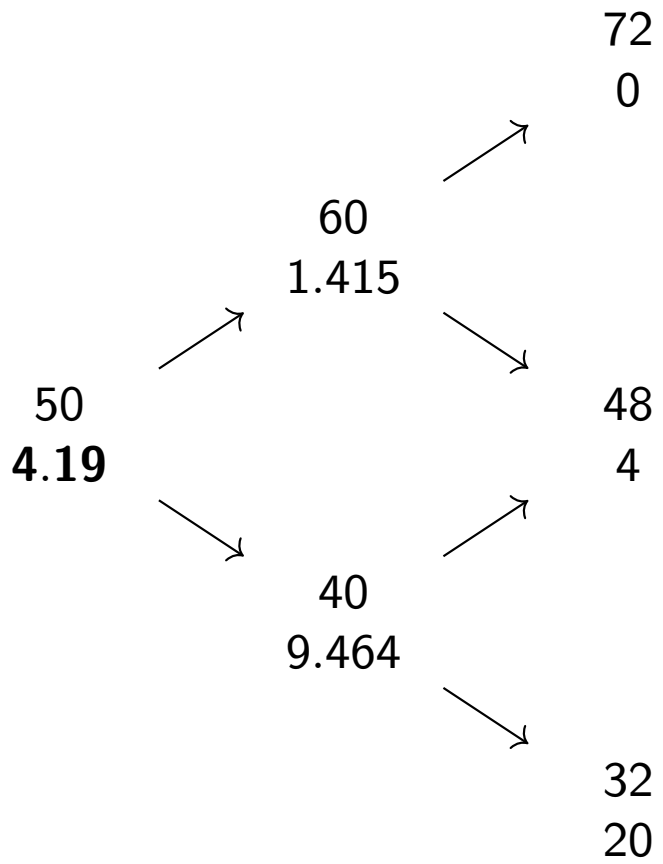
- End nodes: $f_{uu} = 0$, $f_{ud} = 4$, $f_{dd} = 20$.
- Year 1:

  $f_u = e^{-0.05}[\,0.628(0) + 0.372(4)\,] = 1.415$

  $f_d = e^{-0.05}[\,0.628(4) + 0.372(20)\,] = 9.464$
- Today:

  $f = e^{-0.05}[\,0.628(1.415) + 0.372(9.464)\,]$

  $= \boxed{\$4.19}$.



Check with the one-line formula:
$e^{-0.1}[\,2(0.6282)(0.3718)(4) + (0.3718)^2(20)\,] = 4.19.$

The hedge moves — why the ELS desk was sweating

- The replicating portfolio is different at every node. For the put, $x = \dfrac{f_u - f_d}{S_0 u - S_0 d}$ (negative: short stock):

Node	Today ($S = 50$)	Up ($S = 60$)	Down ($S = 40$)
Shares to hold	$\dfrac{1.415 - 9.464}{60 - 40} = -0.40$	$\dfrac{0 - 4}{72 - 48} = -0.17$	$\dfrac{4 - 20}{48 - 32} = -1.00$

- To *replicate* a put you are short stock, and you must **short more as the stock falls**. The ELS issuer is *long* the put (the saver sold it), so its hedge is the opposite: **long** index futures, 0.40 per put today, 1.00 if the index drops.

- March 2020: Korean securities firms were long these puts through about ₩48tn of ELS. Indices fell 30–40% in three weeks. The puts gained *on paper*; the long futures lost *in cash*, with margin calls "exceeding ₩1 trillion for a single day" — and the recipe demanded still more futures, and more margin, the same morning. [KED Global, 25 Mar 2020]

N steps — the end nodes

- N steps of length Δt to expiry; $p = \dfrac{e^{r\Delta t} - d}{u - d}$ at every step.
- Because the tree recombines, there are only $N + 1$ end nodes. Node j is "j ups and $N - j$ downs", with

$$S_T = S_0\, u^j d^{N-j}, \qquad j = 0, 1, \ldots, N.$$

- Many paths lead to node j: any ordering of j ups among N steps. There are $\binom{N}{j} = \dfrac{N!}{j!(N-j)!}$ of them (Excel: `COMBIN(N,j)`).
- Each path has risk-neutral probability $p^j(1-p)^{N-j}$, so node j has probability $\binom{N}{j} p^j (1-p)^{N-j}$.

N steps — the formula

Ups	Downs	RN probability	S_T	Payoff
0	N	$(1-p)^N$	$S_0 d^N$	f_0
1	$N-1$	$N\,p(1-p)^{N-1}$	$S_0 u d^{N-1}$	f_1
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
j	$N-j$	$\binom{N}{j} p^j (1-p)^{N-j}$	$S_0 u^j d^{N-j}$	f_j
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
N	0	p^N	$S_0 u^N$	f_N

- For a **European** option only the end nodes matter, so no backward induction is needed:

$$f = e^{-rT} \sum_{j=0}^{N} \binom{N}{j} p^j (1-p)^{N-j} f_j, \qquad T = N\Delta t.$$

- Discounted risk-neutral expected payoff.

Example — a three-step European call

Q. A 3-year European call has $K = 30$ on a stock at $S_0 = 30$. Each year the stock moves up or down by 10%; $r = 5\%$ (c.c.). Price the call with the N-step formula.

Example — a three-step European call

A. $u = 1.1$, $d = 0.9$, $p = \dfrac{e^{0.05} - 0.9}{0.2} = 0.756$, $N = 3$:

j	Paths $\binom{3}{j}$	RN probability	$S_T = 30(1.1)^j(0.9)^{3-j}$	f_j
0	1	$(0.244)^3 = 0.015$	21.87	0
1	3	$3(0.756)(0.244)^2 = 0.135$	26.73	0
2	3	$3(0.756)^2(0.244) = 0.418$	32.67	2.67
3	1	$(0.756)^3 = 0.433$	39.93	9.93

- $f = e^{-0.05\times 3}\big[\,0.418 \times 2.67 + 0.433 \times 9.93\,\big] = 0.861 \times 5.41 = \boxed{\$4.66}$.
- Drop the "3 paths" at $j = 2$ and you get \$4.02 — the most common wrong answer on this question. Count the paths.

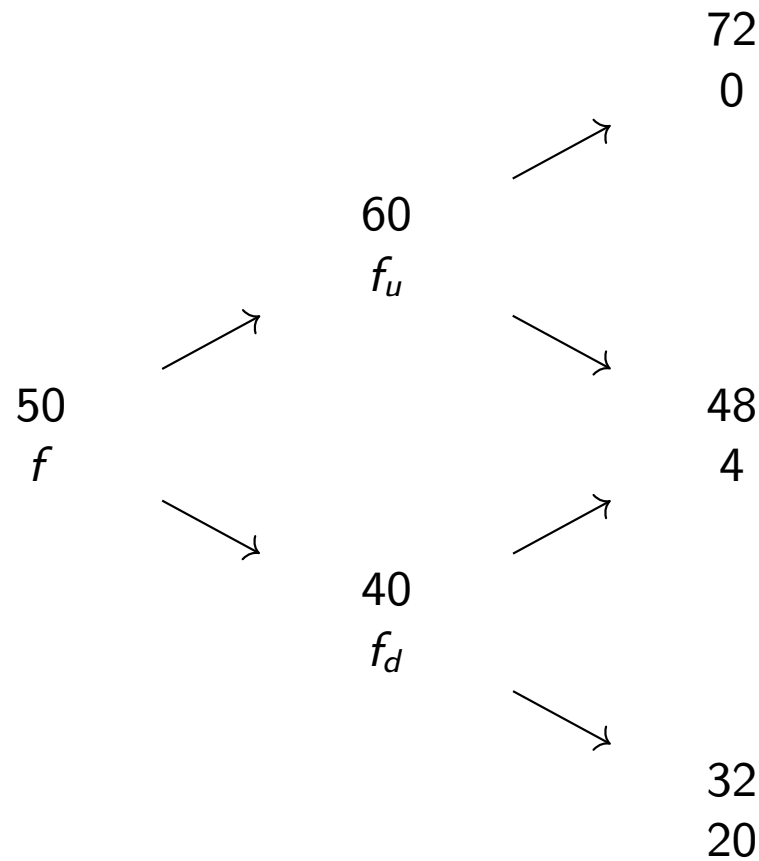
American Options

American options — one extra line at every node

- An American option can be exercised at any node, so at each node the holder compares two numbers:
  1. **Wait:** the value of keeping the option alive, $e^{-r\Delta t}[\,pf_u + (1-p)f_d\,]$;
  2. **Exercise now:** $K - S$ for a put, $S - K$ for a call.
- Option value at the node = the **larger** of the two. Everything else — the tree, p, backward induction — is unchanged.
- This is why trees, not Black–Scholes, are the workhorse for American options: Black–Scholes has no "exercise now" line.

Example — the same put, American

Q. Same tree as before: $S_0 = 50$, $\pm 20\%$ per year, $r = 5\%$. Now the 2-year put with $K = 52$ is **American**. What is it worth, and where is it exercised?



Example — the same put, American

A. $p = 0.6282$ as before. At each year-1 node, wait or exercise:

$$f_u = \max\big(\underbrace{1.415}_{\text{wait}},\ \underbrace{52-60}_{\text{exercise}}\big) = 1.415$$

$$f_d = \max\big(\underbrace{9.464}_{\text{wait}},\ \underbrace{52-40}_{\text{exercise}}\big) = \boxed{12} \quad \leftarrow \text{exercise!}$$

- Today:

$$f = \max\big(e^{-0.05}[\,0.628(1.415) + 0.372(12)\,],\ 52-50\big) = \max(5.09,\ 2) = \boxed{\$5.09}.$$

	European	American	Early-exercise premium
2-year put, $K = 52$	4.19	5.09	0.90

Why exercise at the down node?

- At $S = 40$ the put is deep in the money. Waiting keeps the chance that the stock recovers to 48 (payoff 4) — but it also means waiting a year to collect \$12.
- The interest on \$12 for a year, plus the risk that the stock bounces, outweighs the small extra payoff from waiting. So the holder cashes in. Lecture 7 said American puts *may* be exercised early; the tree shows *when*.
- Run the same tree for an American **call** on a non-dividend stock and the "exercise now" line never wins: $S - K$ is always below the waiting value. The tree confirms Lecture 7's "never exercise early."

Choosing u and d

Where do u and d come from?

- So far u and d were handed to us. In practice we are handed a **volatility** σ: the annualised standard deviation of the stock's return.
- Over a short step Δt the return's standard deviation is $\sigma\sqrt{\Delta t}$ (variances add over time; standard deviations scale with $\sqrt{\text{time}}$).
- Choose u and d so that the tree's one-step return has exactly that standard deviation. The standard (Cox–Ross–Rubinstein) choice:

$$\boxed{u = e^{\sigma\sqrt{\Delta t}}, \qquad d = e^{-\sigma\sqrt{\Delta t}} = 1/u}$$

e.g. $\sigma = 25\%$, $\Delta t = 0.5$: $u = e^{0.25\sqrt{0.5}} = 1.193$, $d = 0.838$. A 6-month step moves the stock $\pm 18\%$.

Why that choice works — the sketch

- Let R be the one-step return in the *real* world (probability $p^* = \frac{e^{\alpha \Delta t} - d}{u - d}$). Its variance is

$$\mathrm{Var}(R) = p^*(u-1)^2 + (1-p^*)(d-1)^2 - \left[e^{\alpha \Delta t} - 1\right]^2 = (u+d)e^{\alpha \Delta t} - ud - e^{2\alpha \Delta t}.$$

- Plug in $u = e^{\sigma\sqrt{\Delta t}}$, $d = e^{-\sigma\sqrt{\Delta t}}$ and expand each exponential to first order in Δt ($e^x \approx 1 + x + x^2/2$):

$$\mathrm{Var}(R) \approx (2 + \sigma^2 \Delta t)(1 + \alpha \Delta t) - 1 - (1 + 2\alpha \Delta t) \approx \sigma^2 \Delta t.$$

- Two things to notice. The tree's volatility is σ, as designed. And α **dropped out**: the volatility calibration does not depend on the expected return — which is why we never needed α in the first place.

The full algebra is in the appendix.

Many steps — where the tree is going

- Take Lecture 8's calendar-spread call: $S_0 = K = 100$, $\sigma = 25\%$, $r = 3\%$, $T = 0.5$.
- Price it on a CRR tree with N steps, each of length $0.5/N$.
- The price bounces, then **settles at \$7.76**.
- That limit is the **Black–Scholes** price. CRR showed the tree converges to it; next week we meet it directly.

Steps N	Call price
1	9.50
2	6.96
5	8.11
10	7.59
20	7.67
50	7.73
100	7.74
500	7.76
∞ (Black–Scholes)	7.76

Other underlyings — change only what the asset grows at

- In the risk-neutral world the *total* return on any traded asset is r. If part of that return is paid out, the *price* grows more slowly, and p must reflect it:

Underlying	Price grows at	Risk-neutral p
Non-dividend stock	r	$\dfrac{e^{r\Delta t} - d}{u - d}$
Stock index paying yield q	$r - q$	$\dfrac{e^{(r-q)\Delta t} - d}{u - d}$
Currency (foreign rate r_f)	$r - r_f$	$\dfrac{e^{(r-r_f)\Delta t} - d}{u - d}$
Futures price	0	$\dfrac{1 - d}{u - d}$

- Futures cost nothing to enter, so there is no capital to earn r on. In the risk-neutral world the futures price is expected to stay where it is: $pF_0u + (1-p)F_0d = F_0$.
- Same u and d from σ; same backward induction. Only p changes.

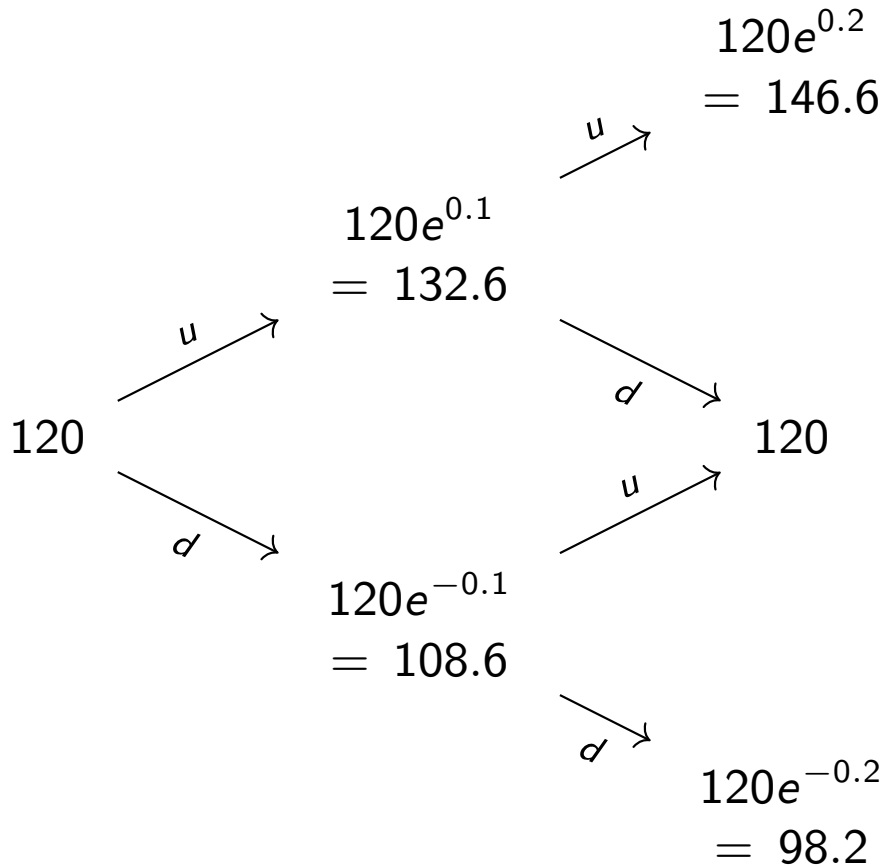
A Detour: Equity as a Call on the Firm

The balance sheet is an option position

- A firm's assets are worth V today and V_T at T, when debt with face value L falls due. V moves like a stock, so put it on a tree.
- At T the creditors get $\min(V_T, L)$ and the shareholders get $\max(V_T - L, 0)$. Assets = Debt + Equity, node by node.
- **Equity is a call on the firm's assets with strike L.** Shareholders have limited liability: if $V_T < L$ they walk away, exactly like an out-of-the-money call.
- **Risky debt is a safe bond minus a put**: $\min(V_T, L) = L - \max(L - V_T, 0)$. Creditors have written the shareholders a put on the firm. [Merton, JF 1974]

Example — valuing equity and debt on a tree

Q1. A firm's assets are worth \$120 million; asset volatility is 10% per annum. Its only debt is a zero-coupon bond repaying \$100 million in two years; $r = 5\%$ (c.c.). Using a two-step tree with $u = e^{\sigma\sqrt{\Delta t}}$, what is the equity worth today?



Example — valuing equity and debt on a tree

A1. $u = e^{0.1} = 1.105$, $d = e^{-0.1} = 0.905$, $p = \dfrac{e^{0.05} - 0.905}{1.105 - 0.905} = 0.731$. Equity payoffs:

$$f_{uu} = \max(146.57 - 100, 0) = 46.57, \quad f_{ud} = 20, \quad f_{dd} = \max(98.25 - 100, 0) = 0.$$

Discount the risk-neutral expectation:

$$E_0 = e^{-0.05\times 2}\left[\,(0.731)^2(46.57) + 2(0.731)(0.269)(20) + 0\,\right] = \boxed{\$29.63 \text{ million}}.$$

Q2. And the debt?

A2. Assets = Debt + Equity, so $D_0 = 120 - 29.63 = \boxed{\$90.37 \text{ million}}$. (Check: discount $\min(V_T, 100)$ on the same tree and you get the same number.)

- The debt promises 100 in two years for 90.37 today: a yield of $\frac{1}{2}\ln(100/90.37) = 5.06\%$, a **credit spread** of 6 basis points over r. With assets at 120 and only 10% volatility, default is a remote tail — the put is nearly worthless.

Turn up the volatility

- Same firm, same debt, but asset volatility 30%: $u = 1.350$, $d = 0.741$, $p = 0.510$. End-node assets 218.6, 120, 65.9.

	$\sigma = 10\%$	$\sigma = 30\%$
Equity	29.63	36.94
Debt	90.37	83.06
Yield on debt	5.06%	9.28%
Credit spread	0.06%	4.28%

- Nothing happened to the assets today. Making them **riskier** moved \$7.3 million from creditors to shareholders — because equity is a call, and calls love volatility.
- This is **asset substitution**, the reason bond covenants restrict what a borrower may do with the money. Option pricing explains a page of your corporate-finance course in one tree.

Summary — what to take away

- An option is priced by **building it**: x shares plus a bond that match its payoff in every state. The price is the cost of the ingredients, and x is the hedge.
- The recipe rearranges into $f = e^{-r\Delta t}[\, pf_u + (1-p)f_d \,]$ with $p = (e^{r\Delta t} - d)/(u - d)$: **discount the risk-neutral expectation at the risk-free rate**. DCF with real probabilities gives the same price, with much more work.
- Many steps: **work backwards** node by node (and count the paths). American: at each node, **max(wait, exercise)**.
- $u = e^{\sigma\sqrt{\Delta t}}$ ties the tree to volatility; as $N \to \infty$ the price converges to Black–Scholes.
- The hedge **changes at every node**. That is the ELS desk's job, and the March 2020 margin calls are what it costs when the index moves fast.

Next week: the tree with infinitely many steps — the Black–Scholes–Merton formula. Where $N(d_1)$ and $N(d_2)$ come from, how to read them, and what changes for indices, futures and currencies.
Reading: Hull, Ch. 15.

Appendix: Why $u = e^{\sigma\sqrt{\Delta t}}$ Matches the Volatility

Appendix — the variance of a one-step return

- One-step gross return: u with real probability p^*, d with $1-p^*$, where $p^* = \frac{e^{\alpha\Delta t}-d}{u-d}$ so that $E[\text{gross return}] = e^{\alpha\Delta t}$.
- Variance of the net return R (same as the gross return's):

$$\begin{aligned}
\text{Var}(R) &= p^* u^2 + (1-p^*)d^2 - \left[p^* u + (1-p^*)d\right]^2 \\
&= p^*(u^2 - d^2) + d^2 - e^{2\alpha\Delta t} \\
&= \frac{e^{\alpha\Delta t} - d}{u-d}(u+d)(u-d) + d^2 - e^{2\alpha\Delta t} \\
&= e^{\alpha\Delta t}(u+d) - d(u+d) + d^2 - e^{2\alpha\Delta t} \\
&= (u+d)\, e^{\alpha\Delta t} - ud - e^{2\alpha\Delta t}.
\end{aligned}$$

Appendix — Taylor expansion

- Taylor series: $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$. For small x keep the first few terms.
- With $u = e^{\sigma\sqrt{\Delta t}}$ and $d = e^{-\sigma\sqrt{\Delta t}}$, so $ud = 1$:

$$u + d \approx \left(1 + \sigma\sqrt{\Delta t} + \tfrac{1}{2}\sigma^2\Delta t\right) + \left(1 - \sigma\sqrt{\Delta t} + \tfrac{1}{2}\sigma^2\Delta t\right) = 2 + \sigma^2\Delta t,$$
$$e^{\alpha\Delta t} \approx 1 + \alpha\Delta t, \qquad e^{2\alpha\Delta t} \approx 1 + 2\alpha\Delta t.$$

- Substitute, dropping terms of order $\Delta t^{3/2}$ and higher:

$$\begin{aligned}\mathrm{Var}(R) &\approx (2 + \sigma^2\Delta t)(1 + \alpha\Delta t) - 1 - (1 + 2\alpha\Delta t) \\ &= 2 + 2\alpha\Delta t + \sigma^2\Delta t + \sigma^2\alpha\Delta t^2 - 2 - 2\alpha\Delta t \;\approx\; \sigma^2\Delta t. \qquad \blacksquare\end{aligned}$$